(iv) Truth table for $(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$

p	q	r	$\neg p$	$(\neg p \rightarrow r)$	$p \leftrightarrow q$	$(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	F	T	F

7. (i) Contradiction (ii) Tautology (iii) Contingency (iv) Tautology

12. $p \rightarrow (q \rightarrow p)$ is a Tautology.

13. Yes. The statements are logically equivalent.

Exercise 12.3

Choose the appropriate answer from the given distractors.

Q	1	2	3	4	5	6	7	8	9	10
A	(2)	(3)	(2)	(4)	(2)	(2)	(3)	(4)	(3)	(2)
Q	11	12	13	14	15	16	17	18	19	20
A	(4)	(1)	(3)	(3)	(3)	(2)	(4)	(3)	(1)	(4)

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